

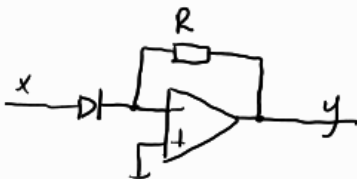
$$x < 0: y = +V_{cc}$$

$$x > 0: \frac{x}{R} = I_D = I_{D0} \left(e^{\frac{U_D}{U_T} - 1} \right) = I_{D0} e^{\frac{y}{U_T}}$$

$U_D = y$ $\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$
 zeh.

$$y = U_T \ln \frac{x}{I_{D0} R}$$

vezje računa $\ln$, ampak je odv. od T , zato ni dober



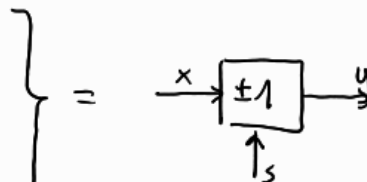
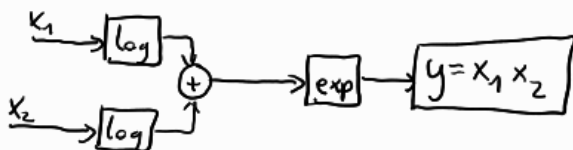
$$x < 0: y = 0$$

$$x > 0: I_D + I_R = 0$$

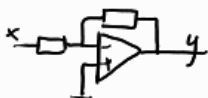
$$I_{D0} (e^{\frac{U_D}{U_T}} - 1) + \frac{y}{R} = 0$$

$$y = -R I_{D0} e^{\frac{x}{U_T}}$$

vezje računa eksponentno funkc.



① S sklenjeno:



$$y = -x$$

② S razklenjeno



$$y = x$$

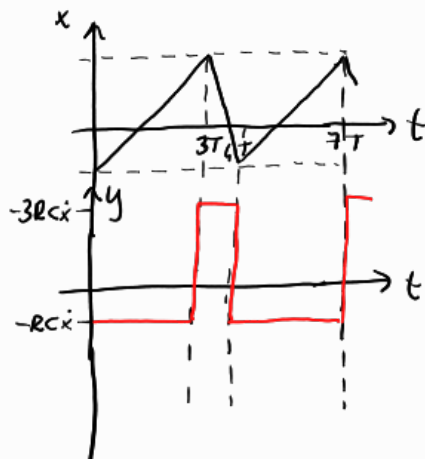
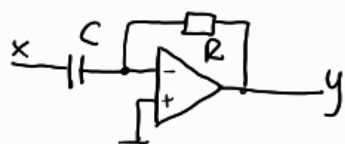
$$T = \frac{y}{x}$$



prenosna funkcija ojačevalnika

- za linearna vezja ok, za nelinearna ni fajna (težko izračunat $\frac{y}{x}$)

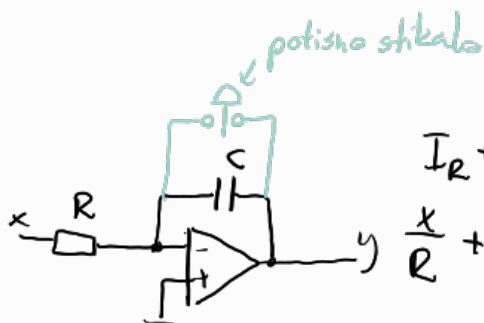
VEZJA S KOND.



$$I_C + I_R = 0$$

$$C \frac{dU_C}{dt} + \frac{y}{R} = C \frac{dx}{dt} + \frac{y}{R} = 0 \rightarrow \boxed{y = -RC \frac{dx}{dt} = -RC \dot{x}}$$

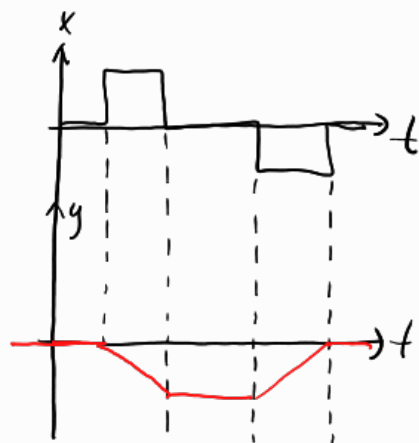
vezje računa odvod



$$I_R + I_C = 0$$

$$\frac{x}{R} + C \frac{dU_C}{dt} = \frac{x}{R} + C \frac{dy}{dt} = 0$$

$$y = -\frac{1}{RC} \int x dt + \text{konst.}$$



če dodamo , je

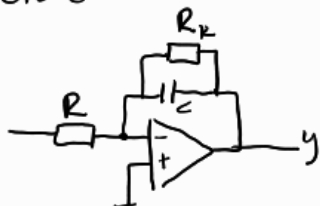
ob stisku (reset)

$$\text{konst.} = 0$$

(do $t=0$ držimo stikalo)

Pomanjkljivost: če vezje pustimo pri miru, bo y počasi rasel zaradi vhodnega toka na operacijskem vezju (nikoli ničelno)

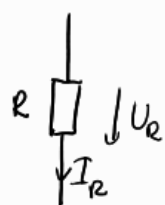
Neidealna rešitev:



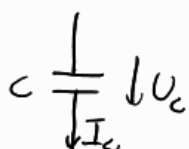
$$\text{Problem: } \frac{x}{R} + C \frac{dy}{dt} + \frac{y}{R_k} = 0$$

Diferencialna enačba

Da bomo lahko računali prenosne funkcije, definiramo "upornosti" elementov

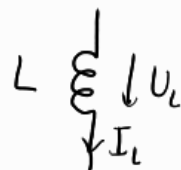


$$I_R = \frac{U_R}{R}$$



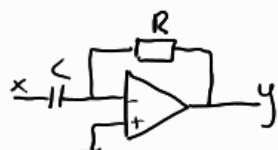
$$I_C = C \frac{dU_C}{dt} ; p = \frac{d}{dt}$$

$$I_C = C_p U_C = \frac{U_C}{\frac{1}{C_p}} = \frac{U_C}{X_C} \leftarrow \text{"upornost" kond.}$$



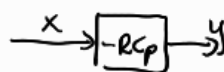
$$U_L = L \frac{dI_L}{dt} = L p I_L$$

$$I_L = \frac{U_L}{L p} = \frac{U_L}{X_L} \leftarrow \text{"upornost" tuljave}$$

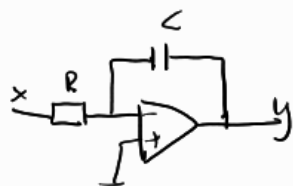


$$I_C + I_R = 0$$

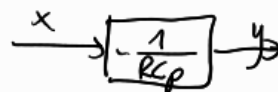
$$\frac{x}{\frac{1}{C_p}} + \frac{y}{R} = 0$$



$$y = -x R C_p \rightarrow T(p) = \frac{y}{x} = -R C_p$$

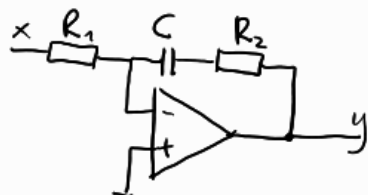


$$\frac{x}{R} + \frac{y}{\frac{1}{C_p}} = 0$$



$$y = -\frac{1}{R C_p} x \rightarrow T(p) = -\frac{1}{R C_p}$$

Ta metoda deluje le za periodične signale!



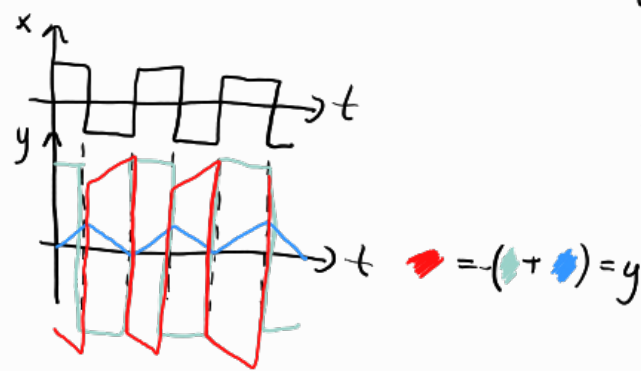
$$\frac{x}{R_1} + \frac{y}{R_2 + \frac{1}{C_p}} = 0$$

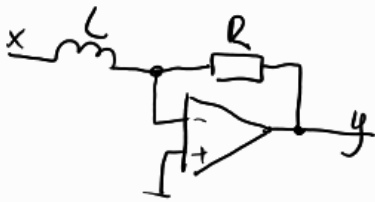
$$-\frac{x}{R_1} = \frac{y C_p}{1 + R_2 C_p}$$

$$y = -x \frac{1 + R_2 C_p}{R_1 C_p}$$

$$T(p) = -\left(\frac{R_2}{R_1} + \frac{1}{R_1 C_p} \right)$$

vezje gačca
vezje integrira
vezje sešteje signala in ju obrne



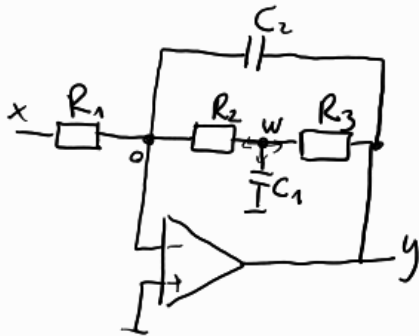


$$I_L + I_R = 0$$

$$\frac{x}{Lp} + \frac{y}{R} = 0$$

$$y = -x \frac{R}{Lp} \rightarrow T(p) = -\frac{R}{Lp} \text{ Integrator}$$

↓
neprava enačba, ker zanemarimo R_T .



$$I_{R1} + I_{R2} + I_{C2} = 0 \rightarrow \frac{x}{R1} + \frac{w-0}{R2} + \frac{y}{\frac{1}{C2p}} = 0$$

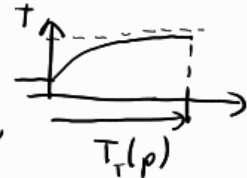
$$I_{R2} + I_{R3} + I_{C1} = 0 \rightarrow \frac{w}{R2} + \frac{w-y}{R3} + \frac{w}{\frac{1}{C1p}} = 0$$

$$x \rightarrow \left[\tau_p \right] \rightarrow \left[\frac{1}{\tau_p} \right] \rightarrow y = x$$

$$x \rightarrow \left[\frac{1}{\tau_p} \right] \rightarrow \left[\tau_p \right] \rightarrow y = x$$

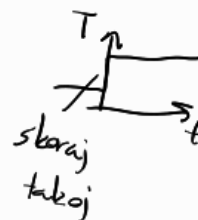
} če poznamo slabosti sistema, lahko signal "predpokravimo", da bo signal na koncu tak, kot ga originalno želimo

↓
npr. termometer dano v krop:

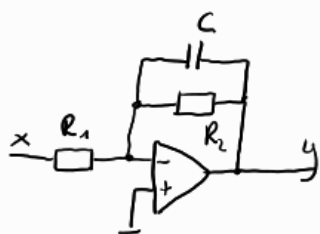


~~$T_T(p)$~~

(počakat je treba veliko časa za natančne meritev)



skraj
takoj



$$xR_2 + yR_1 + yR_1R_2Cp = 0$$

$$yR_1(1 + R_2Cp) = -xR_2$$

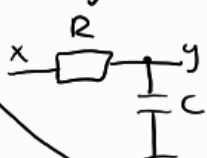
$$y = -x \frac{R_2}{R_1} \frac{1}{1 + R_2Cp}$$

$$T(p) = \frac{y}{x} = - \frac{R_2}{R_1} \frac{1}{1 + R_2Cp}$$

$$I_{R_1} + I_{R_2} + I_C = 0$$

$$\frac{x}{R_1} + \frac{y}{R_2} + \frac{y}{\frac{1}{Cp}} = 0$$

za izračun, kaj vezje počne, najprej pogledamo vezje:



$$y = x \frac{\frac{1}{Cp}}{R + \frac{1}{Cp}} = x \frac{1}{1 + RCp}$$

$$T(p) = \frac{1}{1 + RCp} = \frac{1}{1 + \tau p}$$

analogija s toploto



$$Q = mc_p T$$

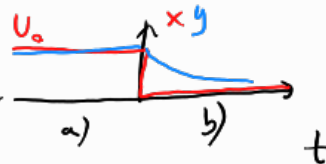
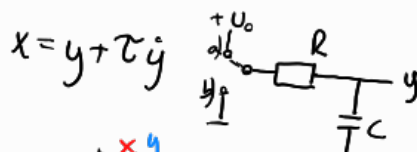
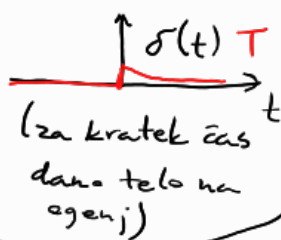
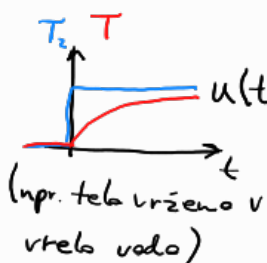
$$\frac{dQ}{dt} = mc_p \frac{dT}{dt}$$

$$j = (T_2 - T)\lambda$$

$$mc_p pT = \lambda T_2 - \lambda T$$

$$T(\lambda + mc_p p) = \lambda T_2$$

$$T = T_2 \frac{1}{1 + \underbrace{\frac{mc_p p}{\lambda}}_{\tau}} = T_2 \frac{1}{1 + \tau p}$$



$$b) y + \tau \dot{y} = 0; y(t=0) = U_0$$

$$y = -\tau \frac{dy}{dt}$$

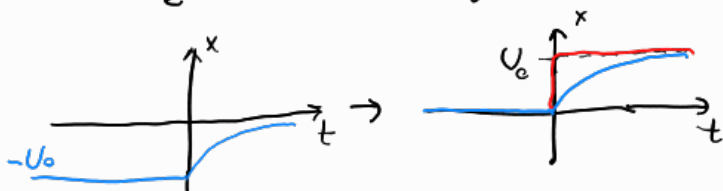
$$-\frac{1}{\tau} dt = \frac{dy}{y} \quad / \int$$

$$\ln k - \frac{t}{\tau} = \ln y / \exp$$

$$y = k e^{-\frac{t}{\tau}}, k = U_0 \text{ (zač. pog.)}$$

$$y = U_0 e^{-\frac{t}{\tau}}$$

linearlost vezja



$$y = U_0 - U_0 e^{-\frac{t}{\tau}} = U_0 (1 - e^{-\frac{t}{\tau}})$$

$$y = U_0 u(t) (1 - e^{-\frac{t}{\tau}})$$

$$U_0 = 1$$

t	y
tau	63%
2tau	86%

$$10\tau \quad 99,995\% \quad \text{Po } 10\tau \text{ je } y=x$$

Vezje za $\tau \gg T$ (amplituda vh. signala)

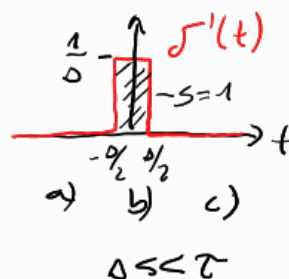
računa povprečje

- za $\tau \approx T$: eksp. približevanje
- za $\tau \ll T$: skoraj hipno eksp. pribl.

$$y + \tau \dot{y} = x$$

$$\tau dy = (x - y) dt \quad (\text{če } y \ll x)$$

$$y = \frac{1}{\tau} \int x dt$$



$$y = U_c$$

$$x = U_R + U_c$$

$$U_R = x - y =$$

$$= U_0 u(t) (1 - (1 - e^{-t/\tau}))$$

$$u(t) = U_0 u(t) e^{-t/\tau}$$

$$= U_0 \delta(t) - U_0 u(t) e^{-t/\tau}$$

$$\delta(t) = U_0 (\delta(t) - u(t) e^{-t/\tau})$$

• če zamenjamo R, c:

$$T(p) = \frac{R}{R + \frac{1}{C}} = \frac{\tau p}{1 + \tau p}$$

- odvod exp. pribl.

$$b) y + \tau \dot{y} = x = \delta'(t)$$

$$\int_{-\infty}^{\infty} y dt + \tau \int_{-\infty}^{\infty} \frac{dy}{dt} dt = \int_{-\infty}^{\infty} \delta'(t) dt$$

pril. lin.

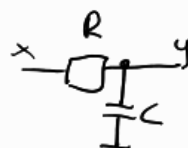
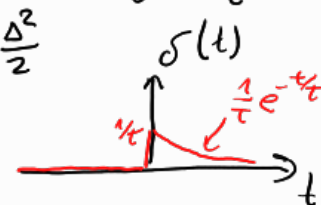
$$y = k(t + \Delta/2)$$

$$\int y dt = k \left(\frac{t^2}{2} + \frac{\Delta t}{2} \right) \Big|_{-\Delta/2}^{\Delta/2} = k \left(\frac{\Delta^2}{8} + \frac{\Delta^2}{4} - \frac{\Delta^2}{8} + \frac{\Delta^2}{4} \right) = k \frac{\Delta^2}{2}$$

$$\lim_{\Delta \rightarrow 0} k \frac{\Delta^2}{2} = 0$$

$$\Rightarrow \tau y(\Delta/2) = 1$$

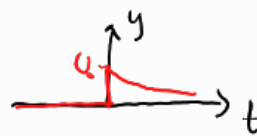
$$y_{kon} = \frac{1}{\tau}$$



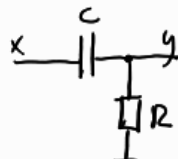
$$T(p) = \frac{1}{1 + \tau p}$$



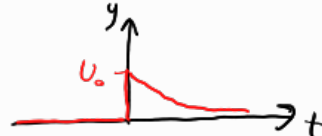
$$y = U_0 u(t) (1 - e^{-t/\tau})$$



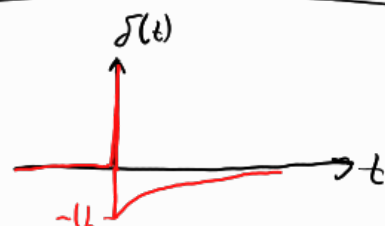
$$y = U_0 u(t) e^{-t/\tau}$$



$$T(p) = \frac{\tau p}{1 + \tau p}$$



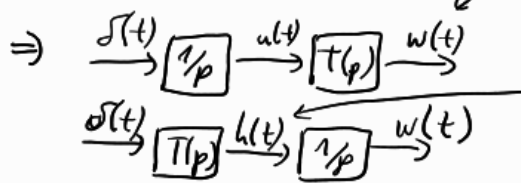
$$y = U_0 u(t) e^{-t/\tau}$$



$$y(t) = U_0 (\delta(t) - u(t) e^{-t/\tau})$$

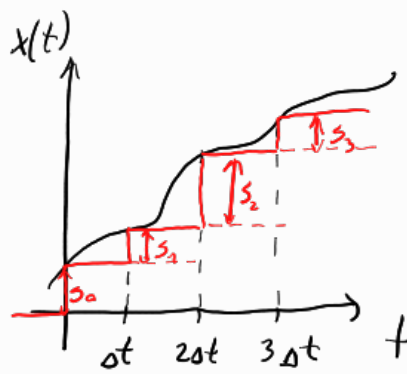
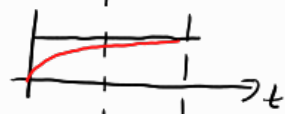
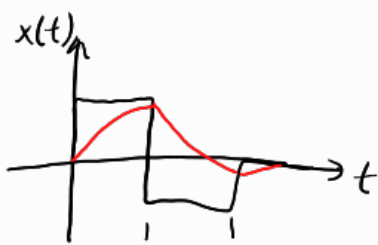


uteznostna funkc. (odziv na nap. stopnice)



$$w(t) = \frac{1}{p} h(t)$$

$$h(t) = p w(t)$$



$$s_n = \left. \frac{dx}{dt} \right|_{t=n\Delta t} \Delta t$$

$$x(t)$$

$$0 < t < \Delta t: x(t) = u(t - 0 \cdot \Delta t) \cdot s_0$$

$$\Delta t \leq t < 2\Delta t: x(t) = s_0 \cdot u(t - 0 \cdot \Delta t) + s_1 \cdot u(t - \Delta t)$$

⋮

$$n\Delta t \leq t < (n+1)\Delta t: x(t) = \sum_n s_n \cdot u(t - n\Delta t)$$

$$y(t) = \sum_n s_n \cdot w(t - n\Delta t) =$$

$$= \sum_n \left. \frac{dx}{dt} \right|_{t=n\Delta t} \cdot w(t - n\Delta t) \Delta t$$

↓ $\lim_{\Delta t \rightarrow 0}$

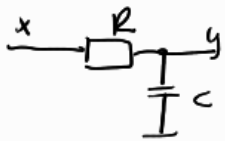
$$y(t) = \int_{-\infty}^t \left. \frac{dx}{dt} \right|_{t=\tau} w(t - \tau) d\tau$$

$$y(t) = \int_{-\infty}^t x(\tau) \cdot w(t - \tau) d\tau$$

Zugled:

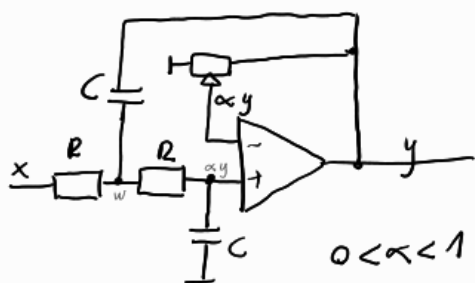


$$x(t) = u(t) \propto t$$



$$w(t) = 1 - e^{-t/\tau_c} \rightarrow y(t) = \int_0^t \left. \frac{dx}{dt} \right|_{t=\tau} w(t - \tau) d\tau = \dots = \alpha t - \alpha R C (1 - e^{-t/\tau_c})$$





w: $I_e + I_c + I_R = 0$

a) $\frac{x-w}{R} + \frac{y-w}{\frac{1}{Cp}} + \frac{\alpha y - w}{R} = 0$

$0 < \alpha < 1$

αy :

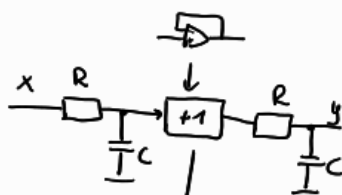
b) $I_e + I_c = 0$

$\frac{\alpha y - w}{R} + \frac{\alpha y}{\frac{1}{Cp}} = 0$

$\Rightarrow y = \frac{1}{\alpha} \frac{1}{1 + T_p(3 - \frac{1}{\alpha}) + T^2 p^2} x$

• če $\alpha = 1$:

$y = x \frac{1}{1 + 2T_p + T^2 p^2} = \frac{x}{(1 + T_p)^2} \rightarrow$



da RC člena ne vplivata en na drugega

Za vezje drugega reda:

$T(p) = \frac{Dp^2 + E_p + F}{Ap^2 + Bp + C} = \frac{y}{x}$

$D\ddot{x} + E\dot{x} + Fx = A\ddot{y} + B\dot{y} + Cy$

• odziv po vzbujanju:

$A\ddot{y} + B\dot{y} + Cy = 0$

$ke^{\alpha t} (A\alpha^2 + B\alpha + C) = 0$

$\begin{cases} y = ke^{\alpha t} \\ \dot{y} = k\alpha e^{\alpha t} \\ \ddot{y} = k\alpha^2 e^{\alpha t} \end{cases}$

$k = 0$: $y = 0$
trivialna rešitev

$k \neq 0$:

$\alpha_{1,2} = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$

$y = k(e^{\alpha_1 t} + e^{\alpha_2 t})$; k določimo iz zač. pog.

1) $B^2 - 4AC > 0$

$\alpha_1, \alpha_2 \in \mathbb{R}$

	$\alpha_1 > 0$	$\alpha_1 < 0$
$\alpha_2 > 0$	ne maramo (hožemo da signal izveni)	ne maramo
$\alpha_2 < 0$	ne maramo	ne maramo

2) ~~$B^2 - 4AC = 0$~~

težko najti R in C ,
da bo točno 0

$Re(\alpha) > 0$



$Re(\alpha) = 0$



$Re(\alpha) < 0$

3) $B^2 - 4AC < 0$

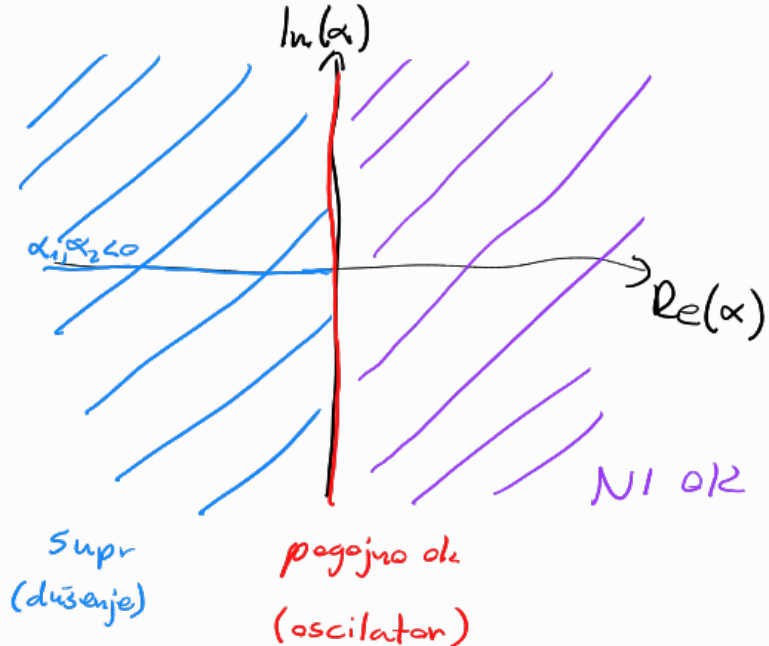
$\alpha_{1,2} = -\frac{B}{2A} \pm i \frac{\sqrt{4AC - B^2}}{2A}$

$= Re[\alpha] \pm i Im[\alpha]$

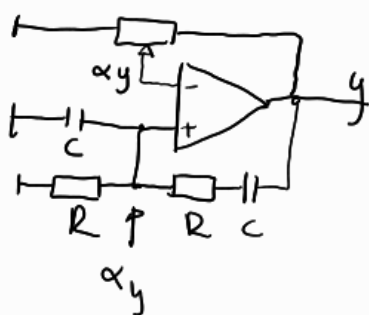
$y = k(e^{Re(\alpha)t} e^{i Im(\alpha)t} + e^{Re(\alpha)t} e^{-i Im(\alpha)t}) =$

$= k e^{Re(\alpha)t} (\cos(Im(\alpha)t) + i \sin(Im(\alpha)t) + \cos(Im(\alpha)t) - i \sin(Im(\alpha)t)) =$

$= k' e^{Re(\alpha)t} \cos(Im(\alpha)t)$



Wienov oscilator



$$I_R + I_C + I_{R_C} = 0$$

$$\frac{\alpha y}{R} + \frac{\alpha y}{\frac{1}{Cp}} + \frac{\alpha y - y}{R + \frac{1}{Cp}} = 0$$

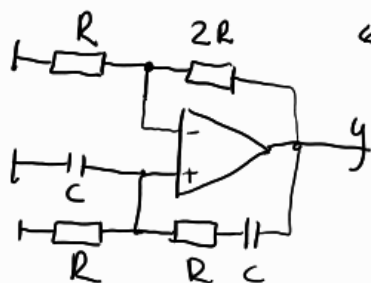


$$\alpha \left(\tau^2 \ddot{y} + \tau \dot{y} \left(3 - \frac{1}{\alpha} \right) + y \right) = 0$$

$$A = \tau^2$$

$$B = \tau \left(3 - \frac{1}{\alpha} \right) = 0 \Rightarrow \alpha = \frac{1}{3}$$

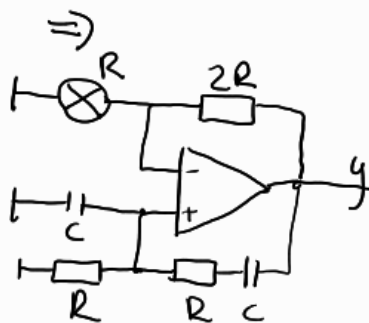
$$C = 1$$



Težko natančno izdelana R, 2R

$$\alpha > \frac{1}{3} \rightarrow B > 0 \Rightarrow \text{growing oscillation}$$

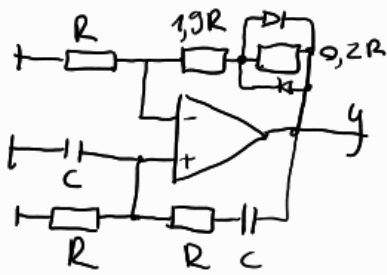
$$\alpha < \frac{1}{3} \rightarrow B < 0 \Rightarrow \text{decaying oscillation}$$



Problem: amplituda se spremeni, če pihnemo v žarnico (jo dodatno hladimo)

Boljše, ker R je odvisna od T

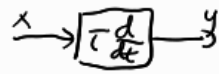
- če y narašča, se žarnica segreje - $R \uparrow \rightarrow \geq$
 - če y pada, se žarnica ohladi - $R \downarrow \rightarrow \leq$
- } vzpostavi se ravnovesje



$$x = A \cos(\omega t + \varphi)$$

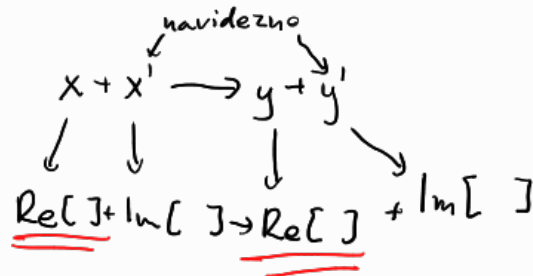
$$y = A' \cos(\omega t + \varphi)$$

$$T(\omega) = \frac{y}{x}$$



$$y = \tau \frac{dx}{dt} = \tau A \frac{d}{dt} \cos \omega t = -\tau A \omega \sin(\omega t)$$

$$T(\omega) = -\frac{\tau \omega \sin \omega t}{\cos \omega t} \quad \text{ni linearno (odv. od } t, \tan x \text{ ni lin.)}$$



$$\rightarrow x = A(\cos \omega t + i \sin \omega t) = A e^{i\omega t}$$

navidezno

za diferenciator:

$$y = A \tau i \omega e^{i\omega t} = A \tau i \omega (\cos \omega t + i \sin \omega t)$$

$$T(i\omega) = \frac{y}{x} = \frac{A \tau i \omega e^{i\omega t}}{A e^{i\omega t} \cdot 1} = i\omega \tau = |T(i\omega)| \cdot e^{i\varphi}$$

$$y = T(i\omega) \cdot x = |T(i\omega)| e^{i\varphi} A e^{i\omega t} = |T(i\omega)| A e^{i(\omega t + \varphi)} = |T(i\omega)| A (\cos(\omega t + \varphi) + i \sin(\omega t + \varphi))$$

fazni zamik

$$\varphi = \arctg \frac{\text{Im}(T(i\omega))}{\text{Re}(T(i\omega))}$$

sprememba
amplitude

$$|T(i\omega)| = \sqrt{T(i\omega) T(-i\omega)}$$

$$T(i\omega) \leftrightarrow T(p):$$

$$T(i\omega) = \underline{\hspace{2cm}}$$

$$T(p) = \frac{Dp^2 + Ep + F}{Ap^2 + Bp + C} = \frac{y}{x} \rightarrow Dp^2x + Ep_x + Fx = Ap^2y + Bpy + Cy$$

$$D\ddot{x} + E\dot{x} + Fx = A\ddot{y} + B\dot{y} + Cy$$

$$x = e^{i\omega t}$$

$$y = T(i\omega)x = T(i\omega)e^{i\omega t}$$

$$\dot{x} = i\omega e^{i\omega t}$$

$$\dot{y} = T(i\omega)i\omega e^{i\omega t}$$

$$\ddot{x} = (i\omega)^2 e^{i\omega t}$$

$$\ddot{y} = T(i\omega)(i\omega)^2 e^{i\omega t}$$

$$D(i\omega)^2 e^{i\omega t} + E i\omega e^{i\omega t} + F e^{i\omega t} = A T(i\omega)(i\omega)^2 e^{i\omega t} + B T(i\omega) i\omega e^{i\omega t} + C T(i\omega) e^{i\omega t}$$

$$T(i\omega) = \frac{D(i\omega)^2 + E i\omega + F}{A(i\omega)^2 + B i\omega + C}$$

$$\Rightarrow \boxed{T(i\omega) = T(p) \Big|_{p \rightarrow i\omega}}$$

$$\frac{1}{C} \quad X_C = \frac{1}{Cp} \rightarrow \underline{X_C(i\omega) = \frac{1}{i\omega C}} \quad \text{impedanca}$$

$$\begin{aligned} \rightarrow X_C = \frac{1}{i\omega C} = \frac{U_C}{I_C} &\rightarrow I_C = U_C i\omega C = U_0 (\cos \omega t + i \sin \omega t) i\omega C = \\ &= U_0 e^{i\omega t} e^{i\frac{\pi}{2}} \omega C = U_0 \omega C e^{i(\omega t + \frac{\pi}{2})} = \\ &= U_0 \omega C \left(\cos(\omega t + \frac{\pi}{2}) + i \sin(\omega t + \frac{\pi}{2}) \right) \end{aligned}$$

↑
Toč prehiteva nap. za 90°

$$T(i\omega) = \frac{D(i\omega)^2 + E(i\omega) + F}{A(i\omega)^2 + B i\omega + C} = \text{konst.} \cdot \frac{(i\omega - \omega_{N1})(i\omega - \omega_{N2})}{(i\omega - \omega_{P1})(i\omega - \omega_{P2})} = \text{konst.} \cdot T_{N1}(i\omega) \cdot T_{N2}(i\omega) \cdot T_{P1}(i\omega) \cdot T_{P2}(i\omega) =$$

$$= \text{konst.} \cdot |T_{N1}| e^{i\varphi_{N1}} |T_{N2}| e^{i\varphi_{N2}} |T_{P1}| e^{i\varphi_{P1}} |T_{P2}| e^{i\varphi_{P2}} = |T| e^{i\varphi}$$

Dobimo lahko le (če polinoma lahko razč. na IR faktorje):

i) konst.

ii) $i\omega\tau$

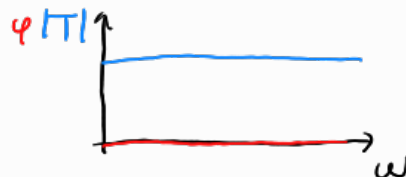
iii) $\frac{1}{i\omega\tau}$

iv) $1 + i\omega\tau$

v) $\frac{1}{1 + i\omega\tau}$

i) $|T(i\omega)| = \text{konst.}$

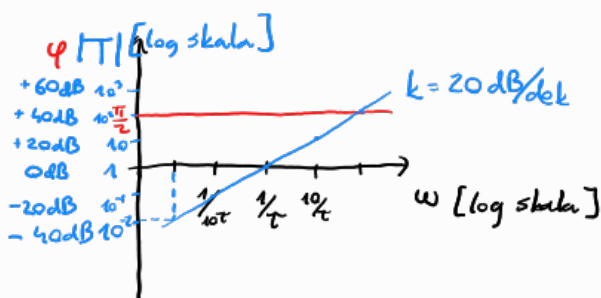
$$\varphi = \arctg \frac{\text{Im}}{\text{Re}} \approx 0$$



ii) $T(i\omega) = i\omega\tau$

$$|T| = \omega\tau$$

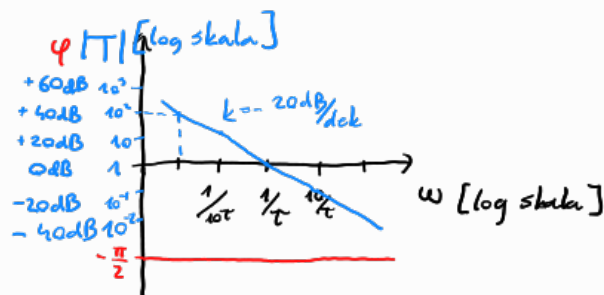
$$\varphi = \arctg \frac{\omega\tau}{0} = \frac{\pi}{2}$$



iii) $T(i\omega) = \frac{1}{i\omega\tau}$

$$|T| = \frac{1}{\omega\tau}$$

$$\varphi = \arctg \frac{\text{Im}}{\text{Re}} = \arctg \left(-\frac{1}{\omega\tau \cdot 0} \right) = -\frac{\pi}{2}$$

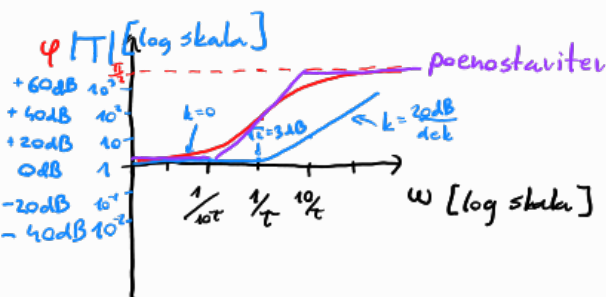


iv) $T(i\omega) = 1 + i\omega\tau$

$$|T| = \sqrt{1 + \omega^2\tau^2}$$

$$\varphi = \arctg \frac{\omega\tau}{1}$$

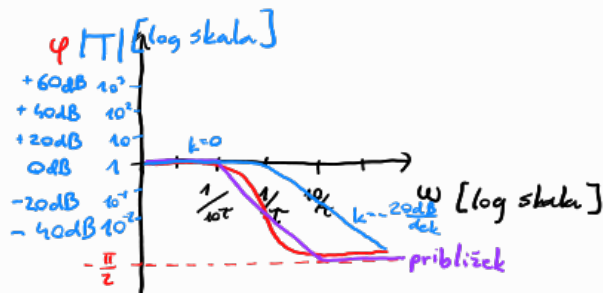
ω	φ
$\frac{1}{100\tau}$	$0,6^\circ$
$\frac{1}{10\tau}$	$5,7^\circ$
$\frac{1}{\tau}$	45°
$\frac{10}{\tau}$	$84,3^\circ$
$\frac{100}{\tau}$	$89,4^\circ$

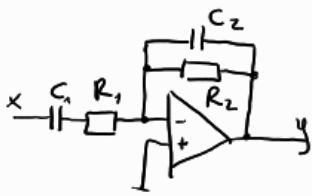


v) $T(i\omega) = \frac{1}{1 + i\omega\tau} = \frac{1 - i\omega\tau}{1 + \omega^2\tau^2}$

$$|T| = \frac{1}{\sqrt{1 + \omega^2\tau^2}}$$

$$\varphi = \arctg \frac{-\omega\tau}{1}$$





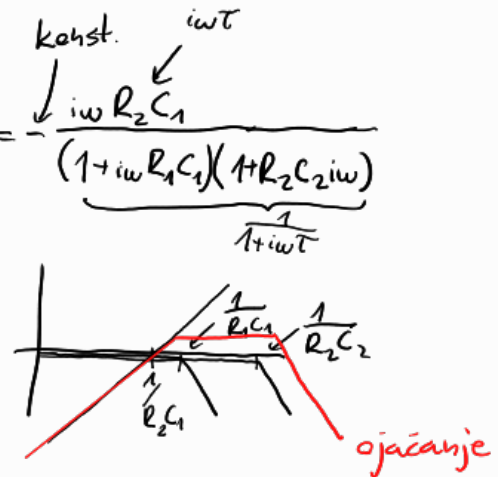
pogledamo limiti:

$$\omega_x \rightarrow 0 \Rightarrow R_c \rightarrow \infty \Rightarrow \omega_y \rightarrow 0$$

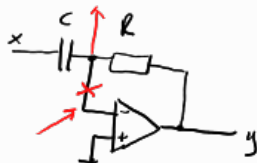
$$\omega_x \rightarrow \infty \Rightarrow R_c \rightarrow 0 \Rightarrow \omega_y \rightarrow 0$$

$$\frac{x}{R_1 + \frac{1}{C_1 p}} + \frac{y}{R_2} + \frac{y}{\frac{1}{C_2 p}} = 0$$

$$T(p) = - \frac{R_2 C_1 p}{(1 + R_1 C_1 p)(1 + R_2 C_2 p)} \rightarrow T(i\omega) = - \frac{i\omega R_2 C_1}{(1 + i\omega R_1 C_1)(1 + R_2 C_2 i\omega)}$$



3 ure manjka

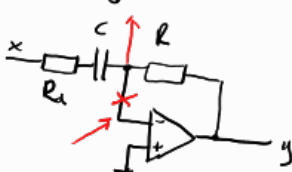


$$T(p) = -\tau_p$$

$$T(i\omega) = -i\omega\tau = i2\pi f\tau$$

$$\uparrow 10^4 \text{ Hz}$$

povečamo stabilnost

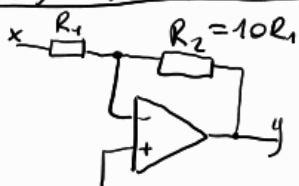


$$R_d = \frac{R}{100}$$

• zmanjša se ojačanje

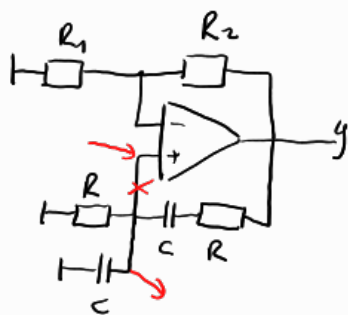
• R_d in C lahko tudi zamenjamo

Hitrejši operacijski ojačevalnik



stabilno za ojačanje > 7

Wienov oscilator



stabilno, če $R_2 = 2R_1$